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Design and Development of a 3D Social Mobile Robot with Artificial Intelligence for Teaching with Assistive Technology

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***Abstract*— In recent years, artificial intelligence is increasingly used in almost every sector of society for the analysis of large databases, pandemic forecasting, image recognition, machine translation, etc. Robots with artificial intelligence, for example, can assist people by performing tasks in extreme or dangerous environments (pandemic, in outer space, in explosive zones, etc.). Robots can also be used in the educational process of children with special educational needs, particularly for teaching children with autism spectrum disorder who may experience difficulties communicating with people. The paper presents the design and development process of a social robot with artificial intelligence for speech and faces emotions recognition. The aim of this paper is to present design process of low-budget 3D prototype of mobile social robot capable of speech recognition, as well as faces emotions recognition. The 3D robot components are designed and printed using 3D software and printing technologies. A seven-step algorithm has been presented for designing and implementation of the robot, ensuring it meets the functional requirements for assistive technology in education. The robot prototype has been implemented and tested with promising results.**

Keywords*— robot, artificial intelligence, assistive technology, special educational needs, autism spectrum disorder

I. INTRODUCTION

In recent years, artificial intelligence is increasingly used in almost all areas of our society for the analysis of large databases, pandemic forecasting, image recognition, machine translation, etc [1]. Projects in the field of artificial intelligence are relevant and at the same time extremely important for the development of our society [2]. Robots with artificial intelligence, for example, can help people to perform actions in extreme conditions, dangerous or inaccessible to human (pandemic, in space, in explosive zones, etc.) [3]. Robots are proving to be assistive and transformative in the educational process for children with special educational needs [4, 5, 6], for example in teaching children with autism spectrum disorder who have difficulty communicating with people [7, 8, 9].

The research focuses on creating a low-budget 3D printed mobile social robot [10] with artificial intelligence capabilities for speech and faces emotions recognition. The robot can be used as a personal assistant for teaching with assistive technology by facilitating communication and interaction in a friendly, engaging manner [11, 12].

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Assistive technologies are designed to support people with special needs to overcome barriers in communication and social interaction. In education, for example, assistive technology allows students to actively develop new skills. Social robots with AI assistive technology, for example, can provide a safe and friendly environment for interaction with children with autism spectrum disorders, who often have difficulty communicating with people [13, 14].

II. ALGORITHM FOR ROBOT DEVELOPMENT

A seven-step algorithm has been proposed for designing a functional prototype of mobile social robot with artificial intelligence capabilities for speech and faces emotions recognition.

A. Choosing a computer mini system

Two Raspberry Pi Zero Wireless computer mini systems [15] with a Header were selected (Fig. 1). The main processor is BCM 2835 SoC with 1 GHz processing speed and 512MB RAM, with included Wi-Fi and Bluetooth for wireless connection.

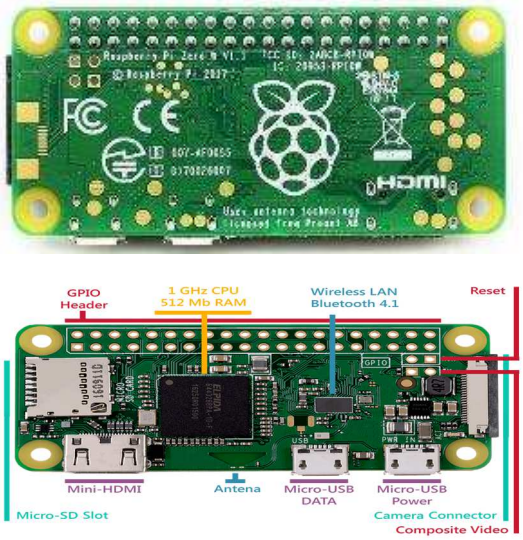


Fig. 1. Minicomputer system

B. Selecting components for face emotions recognition

For face emotions recognition, a Vision Bonnet expansion board was paired with a Raspberry Pi version 2 camera with 8 MP high resolution image capture for accurate recognition (Fig. 2) [16].

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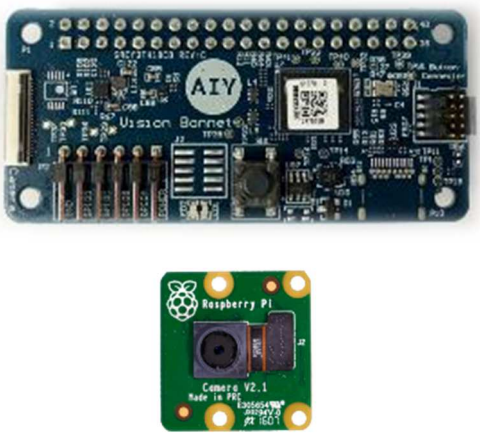


Fig. 2. A board for face emotions recognition and camera

C. Selecting components for speech recognition

A Voice Bonnet expansion board with two built-in microphones and an external speaker is shown for speech recognition (Fig. 3) [17].



Fig.3. A board for speech recognition and speaker

D. Selection of robot's movements mechanisms

One Lego Interactive Servo Motor for each leg and one Lego Medium Servo Motor for the head were chosen for their precision and modularity to drive the robot. Lego EV3 Intelligent Brick was chosen to manage and program the robot's movements. The components are shown on Fig.4.



Fig. 4. Components for driving the robot

E. 3D modelling and printing

3D models were designed using SolidWorks software. An original ergonomic design of robot body has been designed to minimize material usage. The robot has a child-friendly appearance with thin body in which it will be possible to insert all components. Fig. 5 shows 3D CAD design model of the legs and the attached drive motors and controller. Fig. 6 shows 3D CAD design model of robot body with assembled sensors package. Fig. 8 shows 3D CAD design model of overall robot appearance with assembled main housing.

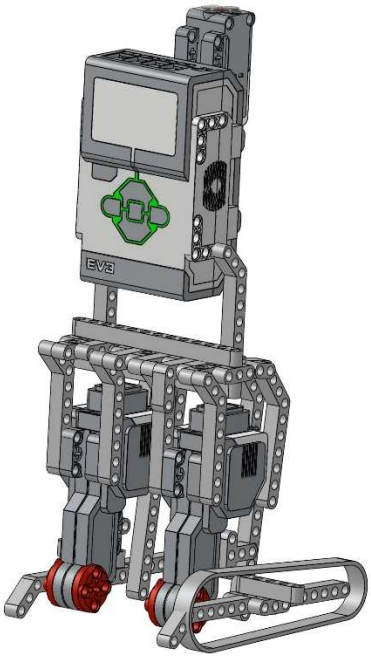


Fig. 5. 3D CAD design model of the legs and the attached drive motors and controller (Drive Assembly)

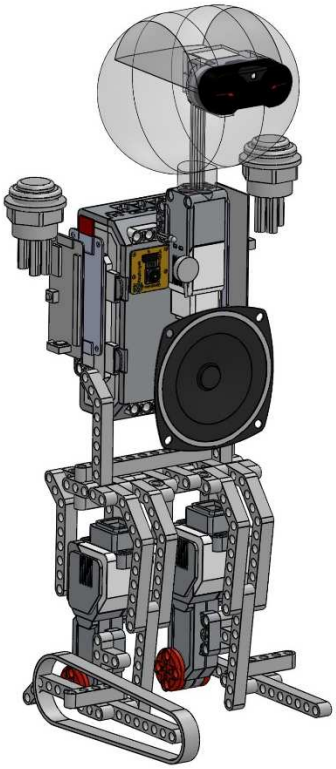


Fig. 6. 3D CAD design model of robot body with assembled sensors package

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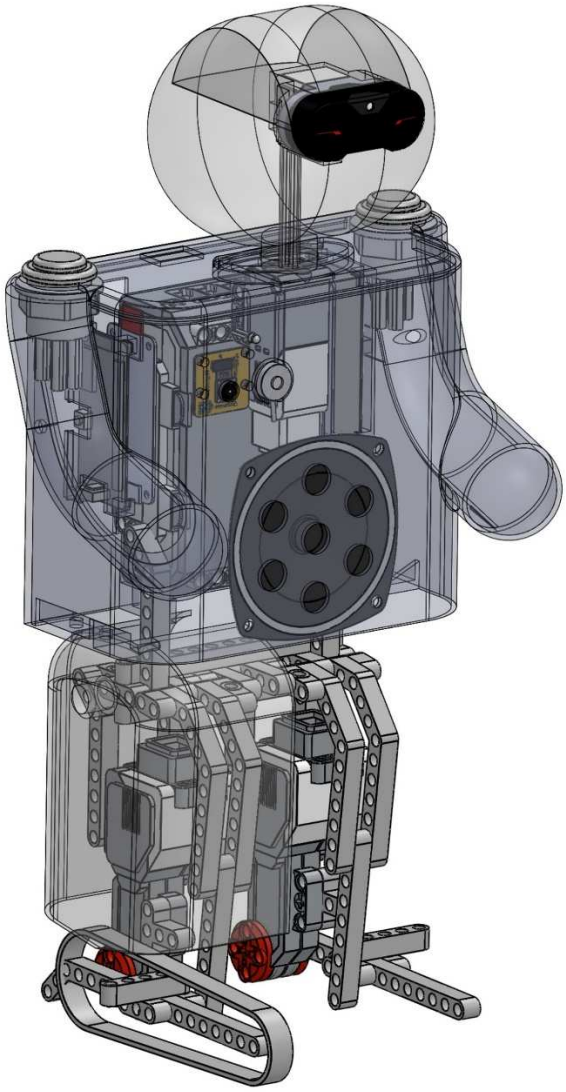


Fig. 8. 3D CAD design model of overall robot appearance with assembled main housing

F. *Printing the plastic parts of the robot on a 3D printer*

For printing the plastic parts, coloured plastic material is used - filament 1.75 mm thickness in blue glossy colour. The parts of the robot can be printed in other colours, and more expensive materials are available, which imitate metal. The blue colour was chosen to please children with special needs, and they aim to perceive the robot as a colourful friendly toy, to relax and communicate with him. The 3D printer Anycubic Mega S is shown on Fig. 9. It is low budget printer and can be purchased at a price of about BGN 400 to be used at home for printing various household items and children's games.

G. *Assembling the robot*

All components used to build the robot are shown on Fig.10. In the process of assembling the robot some specialized tools for measuring and connecting are used: calliper, screwdriver, wrenches, fasteners and more, ensuring the assembled robot met the specified functional criteria.

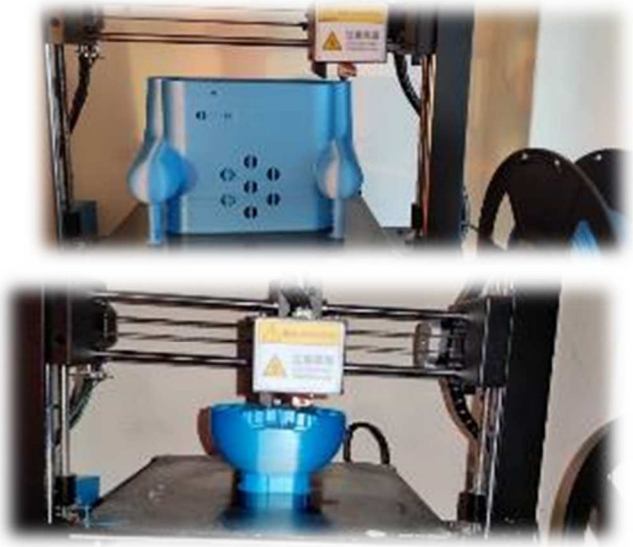


Fig. 9. Printing robot's parts on a 3D printer



Fig. 10. Robot's components

H. *INSTALLING AND PROGRAMMING A PROTOTYPE OF A ROBOT*

Fig. 11 and Fig. 12 show the robot prototype.

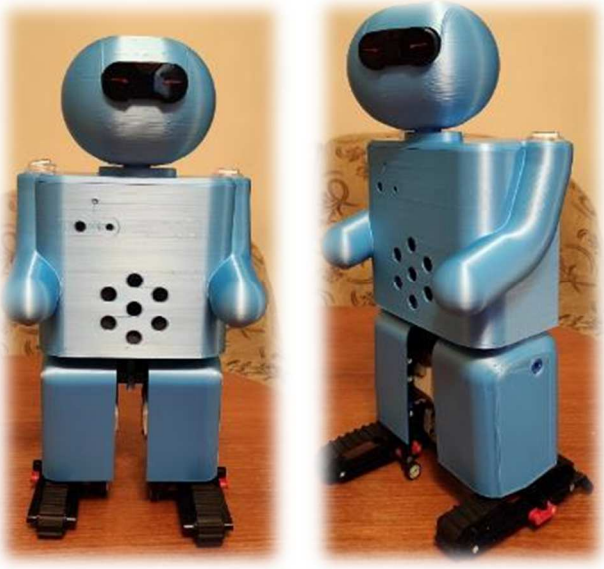


Fig. 11. Robot prototype – front

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Fig. 12. Robot prototype – back

Fig. 13 shows an algorithm for moving the legs:

- Walking type movements;
- Rotation in place;
- and others.

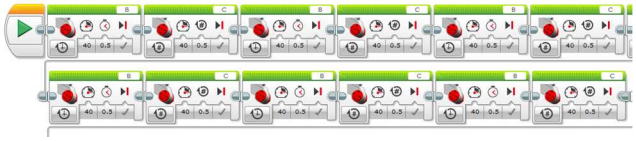


Fig.13. An algorithm for moving robot's legs

Raspbian Linux operating system is installed on minicomputer systems. The AIY Vision Kit software was used to test the ability to recognize faces emotions. Fig.14 presents stages for recognizing face emotions and the software.

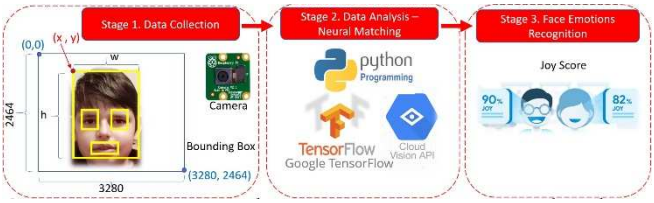


Fig. 14. Stages for face emotions recognition

The main task that is solved for image recognition is the segmentation task in order to emphasize the individual objects in the face (in this case, the eyes and mouth are recognized). The segmentation task is simplified, as the image is considered as a set of pixels - a set of binary values, as shown on Fig. 14.

Fig. 15 presents the architecture with hardware and software for face emotions recognition.

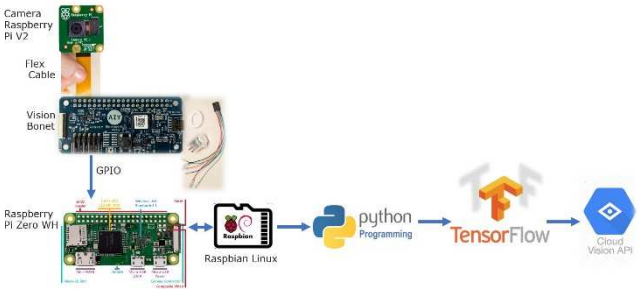


Fig. 15. Architecture with hardware and software for face emotions recognition

Fig. 16 presents the architecture with hardware and software selected for speech recognition.

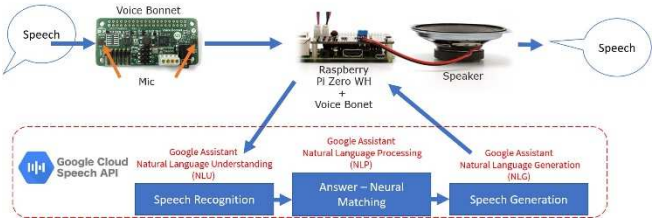


Fig. 16. Architecture with hardware and software solutions selected for speech recognition

III. EXPERIMENTS AND RESULTS

Extensive testing was conducted to evaluate the robot's capabilities in speech recognition, moving and interaction.

The results of testing the movements of the prototype proved that the robot demonstrate stable and reliable movement navigating on various terrains without falling. The results allow us to conclude that the possibilities of robot movement can be expanded with possibility of carrying small loads - for example delivering items or medicines to ill patients.

At this stage, the robot can recognize some facial expressions with more accuracy on happy face. The robot has signals for the different emotions by lighting and switching off a specially mounted light indicator, shown on Fig. 17.

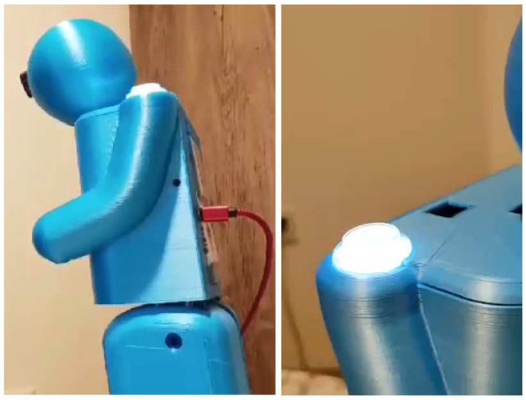


Fig. 17. Light indicator for face emotions recognition

The robot can move also its head left and right using a programmable electrical motor.

Fig. 18 shows a test of the movements of the robot's legs. Both legs are controlled by two programmable electrical motors. The robot moves with the help of rubber chains driven by electrical motors. The legs are separated from the body to allow "walking-type" movement - sequential movement of

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one leg and the other. The rotation in place shown on Fig. 18, is done by programming the motors in reverse directions for speed.



Fig.18. Test of the movements of the robot's legs

The results of testing the ability to recognize speech and communicate with the robot proved that the robot could accurately interpret the meaning of questions and instructions and can respond a meaningful speech answer. The robot can also execute simple commands (such as playing music and etc.). The robot communicates at this stage in English and Bulgarian. Fig. 19 presents a fragment of the developed Python program code in PyCharm program for a robot control library with voice commands in Bulgarian.

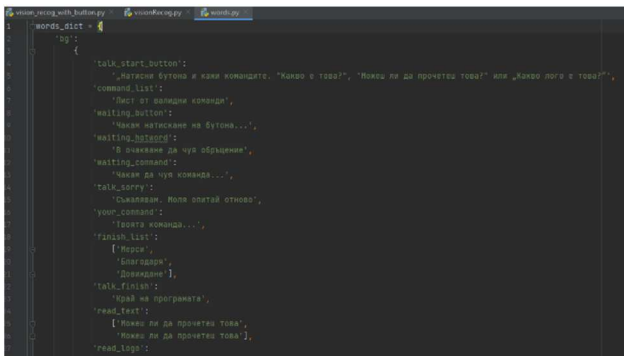


Fig. 19. Python program code in PyCharm program for a robot control library with voice commands in Bulgarian

The program code is optimized for easy integration and customization, which allows for additional adaptations to the specific user needs.

IV. METHODOLOGY FOR TESTING WITH ASD CHILDREN

The robot prototype was tested with children with autism spectrum disorders, who used a limited set of words. The interaction was conducted in an environment that encouraged calm communication, which is especially important for children with sensory sensitivities. The robot successfully interpreted the meaning of words and communicate by providing meaningful voice responses. In addition, robot performed simple commands, such as playing music. This confirms its applicability as a training tool. The results showed that the robot can support the learning process by providing an interaction.

The experiment focused on two key aspects: how the robot understands spoken words and how accurately it can recognize emotions.

- **Participants and testing environment:** 5 children between the ages of 7 and 12 participated in the experiments.
- **Environment:** Testing took place in a quiet room to minimize distractions.
- **Interaction Session Time:** The children engaged in a 10-15 minutes interactive session with the robot.
- **Procedure for Emotion Recognition:** For emotion recognition, children were asked to make five different facial expressions. The children were presented with a series of 5 images of different emotions, which they had to express while the robot tried to recognize them. Each child was asked to display five different facial expressions (neutral, happy, sad, surprised, and angry). The robot achieved 82% accuracy in recognizing emotions, with better performance on happy expressions, Fig. 20. Children with autism often display emotions in ways that are less typical, making emotion recognition difficult. Happiness was the most easily recognized emotion in this study. Future research and development of emotion recognition systems should prioritize improving the detection of other emotions to better support the emotional well-being of children with autism.

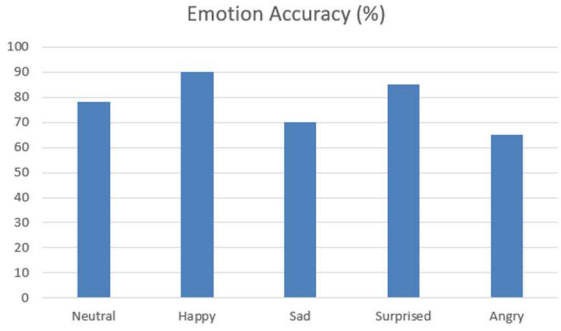


Fig. 20. Emotion Accuracy Robot Recognition Results

- **Procedure for speech recognition:** Voice commands, included simple instructions with simple words, were used: start, stop, say hello, say cat, say dog. Speech recognition accuracy reached 75% (Fig. 21), and the robot responded to commands in about 2,5 seconds on average (Fig. 22).

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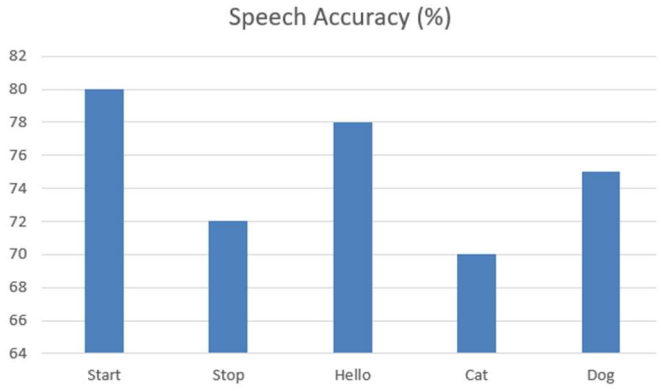


Fig. 21. Accuracy of Speech Recognition Results

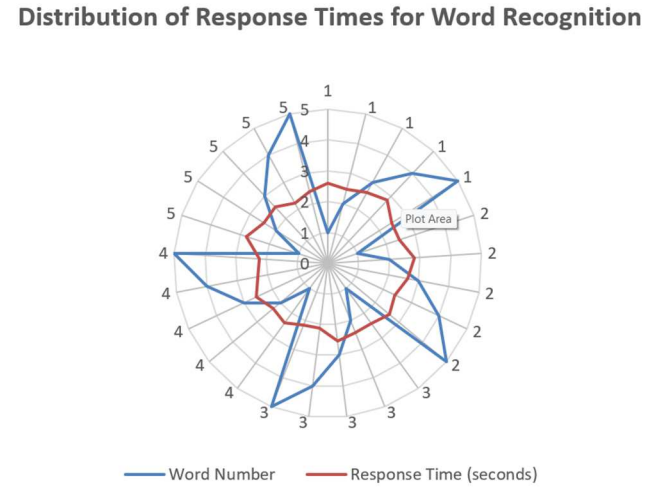


Fig. 22. Results for Distribution of Response Times for Word Recognition

V. CONCLUSIONS

Presented 3D printed robot prototype is a low budget solution build with prefabricated parts at affordable price. A low-budget 3D printer with affordable material is used to print the parts of the robot, which simplifies the process of making specific plastic parts. The use of affordable components and materials makes it applicable in various educational and therapeutic environments, especially when working with children with special educational needs, such as those with autism spectrum disorders.

Tests with children with autism spectrum disorders have shown that the robot helps improve their communication and social skills. Robot's design, which includes bright colours, makes it attractive and engaging for children.

The 3D robot design and programmable system architecture allow for easy customization and upgrade of the robot's functionality adding new modules.

The created prototype of a social robot with artificial intelligence can be integrated into various real educational scenarios for adaptive learning and support of children with different special educational needs.

Despite the positive results, further research is needed to improve the emotion and speech recognition algorithms. Additionally, future tests could include a wider range of users with different needs to increase the reach of the project.

ACKNOWLEDGMENT

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